

CS3 Hook Document

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Have you ever stepped outside and felt like the air was “bad”? Or held your breath when a diesel truck drove by? You’re not wrong to do so – air quality has been linked to numerous health effects and one main pollutant to watch out for is PM2.5. These are liquid and solid particle matter less than 2.5 micrometers in diameter released by industrial emissions. If you’re interested in living in a concrete jungle like New York specifically, you may want to consider which community district carefully because PM2.5 levels can vary across locations. If you’re having a bad air quality day where you live, what about your friend in a more crowded neighborhood two blocks away?

In a bustling city, it’s important to understand how each group is affected. With this report, you can be the person to look through the patterns of air quality over time to see how different communities differ in their measurements. Our scripts and sources invite you to familiarize yourself with the differences over time in PM2.5 levels across New York’s most urban and rural community districts (and their populations), using STL decomposition to examine different trends and seasonal patterns.

Beginning with the script `variousMaterials/scripts/data_cleaning_visualizations.ipynb`, you will explore the data csv file. This will include filtering, conversions, and exploratory plots. From there, the analysis script `variousMaterials/scripts/performed_analysis.ipynb` will guide the main analysis workflow across community districts. According to the Mann-Whitney U test, there will be no significant effect of population levels on PM2.5 in communities, but each step reveals how time series comparisons are made and the patterns taken into account, and may give you a closer look into where they could be lacking.

Time series techniques and statistical testing will evaluate observed differences and you will interpret whether there is room for doubt. Your task is to use our repository, engage with the data and analysis, and learn about how time series data can reveal patterns across community districts related to population and PM2.5.

GitHub Repository: https://github.com/kellykohout/DS4002_CS3/tree/main